

**Ex. 44.** In a division sum, the divisor is ten times the quotient and five times the remainder. If the remainder is 46, determine the dividend.

**Sol.** Remainder = 46 ; Divisor =  $5 \times 46 = 230$  ; Quotient =  $\frac{230}{10} = 23$ .

$\therefore$  Dividend = Divisor  $\times$  Quotient + Remainder =  $230 \times 23 + 46 = 5336$ .

**Ex. 45.** If three times the larger of the two numbers is divided by the smaller one, we get 4 as quotient and 3 as remainder. Also, if seven times the smaller number is divided by the larger one, we get 5 as quotient and 1 as remainder. Find the numbers.

**Sol.** Let the larger number be  $x$  and the smaller number be  $y$ .

Then,  $3x = 4y + 3 \Rightarrow 3x - 4y = 3$  ...(i)

And,  $7y = 5x + 1 \Rightarrow -5x + 7y = 1$  ...(ii)

Multiplying (i) by 5 and (ii) by 3, we get:

$15x - 20y = 15$  ...(iii) and  $-15x + 21y = 3$  ...(iv)

Adding (iii) and (iv), we get:  $y = 18$ .

Putting  $y = 18$  in (i), we get:  $x = 25$ .

Hence, the numbers are 25 and 18.

**Ex. 46.** A number when divided by 6 leaves remainder 3. When the square of the same number is divided by 6, find the remainder.

**Sol.** On dividing the given number by 6, let  $k$  be the quotient and 3 the remainder.

Then, number =  $6k + 3$ .

Square of the number =  $(6k + 3)^2 = 36k^2 + 9 + 36k = 36k^2 + 36k + 6 + 3$   
 $= 6(6k^2 + 6k + 1) + 3$ , which gives a remainder 3 when divided by 6.

**Ex. 47.** Find the remainder when  $9^6 + 7$  is divided by 8.

**Sol.**  $(x^n - a^n)$  is divisible by  $(x - a)$  for all values of  $n$ .

So,  $(9^6 - 1)$  is divisible by  $(9 - 1)$ , i.e.  $8 \Rightarrow (9^6 - 1) + 8$  is divisible by 8  $\Rightarrow (9^6 + 7)$  is divisible by 8.

Hence, required remainder = 0.

**Ex. 48.** Find the remainder when  $(397)^{3589} + 5$  is divided by 398.

**Sol.**  $(x^n + a^n)$  is divisible by  $(x + a)$  for all odd values of  $n$ .

So,  $[(397)^{3589} + 1]$  is divisible by  $(397 + 1)$ , i.e. 398

$\Rightarrow [(397)^{3589} + 1] + 4$  gives remainder 4 when divided by 398

$\Rightarrow [(397)^{3589} + 5]$  gives remainder 4 when divided by 398.

**Ex. 49.** If  $7^{126}$  is divided by 48, find the remainder.

**Sol.**  $7^{126} = (7^2)^{63} = (49)^{63}$ .

Now, since  $(x^n - a^n)$  is divisible by  $(x - a)$  for all values of  $n$ ,

so  $[(49)^{63} - 1]$  or  $(7^{126} - 1)$  is divisible by  $(49 - 1)$  i.e. 48.

$\therefore$  Remainder obtained when  $(7)^{126}$  is divided by 48 = 1.

**Ex. 50.** Find the remainder when  $(257^{166} - 243^{166})$  is divided by 500.

**Sol.**  $(x^n - a^n)$  is divisible by  $(x + a)$  for all even values of  $n$ .

$\therefore (257^{166} - 243^{166})$  is divisible by  $(257 + 243)$ , i.e. 500.

Hence, required remainder = 0.

**Ex. 51.** Find a common factor of  $(127^{127} + 97^{127})$  and  $(127^{97} + 97^{97})$ .

**Sol.**  $(x^n + a^n)$  is divisible by  $(x + a)$  for all odd values of  $n$ .

$\therefore (127^{127} + 97^{127})$  as well as  $(127^{97} + 97^{97})$  is divisible by  $(127 + 97)$ , i.e. 224.

Hence, required common factor = 224.

**Ex. 52.** A 99-digit number is formed by writing the first 59 natural numbers one after the other as:

1234567891011121314.....5859

Find the remainder obtained when the above number is divided by 16.

**Sol.** The required remainder is the same as that obtained on dividing the number formed by the last four digits i.e. 5859 by 16, which is 3.